

AURA (Adaptive Universal Robotic Assistant)

AURA is a mobile AI-powered IoT guardian combining home automation, emotional interaction, and active safety.

Core Innovation: Predictive, autonomous, and emotionally aware system that observes, understands, and acts.

Key Capabilities

- Emotional interaction (face + voice analysis)
- Smart home control (like Alexa/Siri but mobile)
- Safety: fire, smoke, gas, water leakage detection
- Elderly care: fall detection, monitoring
- Child care: cry detection
- Autonomous patrol with live video
- Environmental monitoring (CO2, temperature, humidity)

System Architecture

Hardware: Raspberry Pi/ESP32, sensors, camera, motors, GSM module

Network: WiFi + GSM, MQTT/HTTP

Cloud: AI processing, storage, alerts

Application: mobile app dashboard

Functional Requirements

- Detect hazards and human conditions
- Send alerts via SMS/call/app
- Control appliances
- Navigate indoor environment

Non-Functional Requirements

- Low latency, reliability, security, energy efficiency, scalability

Vision

AURA is a living, protective presence inside your home.